
What is AWS Elastic Beanstalk?

AWS Elastic Beanstalk is a **Platform as a Service (PaaS)** that makes it easy to deploy, manage, and scale web applications and services on AWS. It automatically handles the **infrastructure, deployment, load balancing, auto scaling, and monitoring**—so you can focus on writing code.

Key Features

Simplified Deployment

- Upload your **ZIP/WAR/Docker** app and Beanstalk does the rest.
- Supports deployment through **AWS Console, CLI, or APIs**.

Supported Languages & Platforms

- Languages: **Java, Python, Node.js, .NET, PHP, Ruby, Go**
- Web servers: **Apache, Nginx, IIS**
- Container support: **Docker (single or multi-container)**

Managed Infrastructure

- Auto-provisions: **EC2, Load Balancer, Auto Scaling, RDS (optional)**
- **CloudWatch** for built-in monitoring and alarms

Environment Types

- **Web Server Environment**  – Handles HTTP requests

- **Worker Environment** ⚙️ – Background task processing using SQS
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Architecture Components

Component	Description
 Application	Logical collection of app versions
 App Version	Specific deployable version of your code
 Environment	Deployment target with AWS resources
 Config Settings	Instance type, scaling rules, etc.

How Deployment Works

1.  **Create an Application**
 2.  **Upload Code** (ZIP, WAR, or Docker image)
 3.  **Beanstalk Provisions Resources**
 4.  **Monitor & Manage** via Console/CLI
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Scaling & Monitoring

-  **Auto Scaling** adjusts EC2 instances based on traffic
-  **CloudWatch** monitors:
 - CPU usage

- Latency
 - Request count
 -  **Health checks** and environment status dashboards
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Customization with **.ebextensions**

- Modify software settings
- Install packages
- Set environment variables
- Define custom commands and scripts

Example:

packages:

yum:

git: []

Security Features

-  **IAM roles** for permissions
-  **HTTPS** & custom domain support
-  **Secure access** with Security Groups, VPC
-  Use **Secrets Manager/Parameter Store** for credentials

Benefits of Elastic Beanstalk

Feature

Benefit

 Fully managed	No need to manage EC2, load balancers, etc.
 Fast deployment	Focus on code, not infrastructure
 Scalable	Automatically scales with traffic
 Built-in monitoring	Integrated with CloudWatch
 CI/CD integration	Works with CodePipeline, GitHub Actions, etc.
 Cost-effective	Pay only for underlying resources (EC2, ELB, etc.)

Use Cases

-  Hosting **eCommerce sites, blogs, and APIs**
 -  Testing **MVPs** and prototypes quickly
 -  Deploying **microservices** using Docker
 -  Backend services for **mobile/web apps**
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Alternatives

Service

Description

 AWS Lambda	Serverless computing (no infrastructure to manage)
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 ECS / EKS	Full control over container orchestration
 EC2 + CloudFormation	For advanced customization & control
 AWS App Runner	Simple PaaS for containers or source code

✓ When to Use Beanstalk

- ✓ You want **fast deployment** without managing servers
 - ✓ You use **supported languages** like Node.js, Python, Java
 - ✓ Your app needs to **auto-scale**
 - ✓ You need some **customization**, but not full DevOps control
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